



Solution Set

Question 1

$$\|a\| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14}$$

The unit vector is

$$\hat{a} = \frac{1}{\sqrt{14}}(2, 3, 1)$$

Question 2

$$a \cdot b = 3(-4) + 4(3) = 0$$

The vectors are perpendicular, so

$$\theta = \frac{\pi}{2} = 90^\circ$$

Question 3

Because

$$b = (6, 8) = 2(3, 4) = 2a$$

a is already in the direction of b. The vector projection is

$$\text{proj}_b(a) = \frac{a \cdot b}{b \cdot b} b = \frac{50}{100} b = \frac{50}{100} (6, 8) = (3, 4)$$

Question 4

Perpendicular means dot product zero:

$$(c, 1, 1) \cdot (c, 2c, 1) = c^2 + 2c + 1 = (c + 1)^2 = 0$$

so

$$c = -1$$

Question 5

$$\|a\| = \sqrt{5^2 + 12^2} = 13$$

The two perpendicular scalar projections form legs of a right triangle with hypotenuse 13. If one projection is 7, the other is

$$\sqrt{13^2 - 7^2} = \sqrt{120} = 2\sqrt{30}$$